

```
# Install required libraries
!pip install chromadb sentence-transformers

# Import libraries
import chromadb
from sentence_transformers import SentenceTransformer

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sen
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kuberne
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from requests-oauthlib->kuberne
Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (22.4 MB)
 22.4/22.4 MB 55.8 MB/s eta 0:00:00
Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
 278.2/278.2 kB 21.6 MB/s eta 0:00:00
Downloading build-1.4.2-py3-none-any.whl (24 kB)
Downloading kubernetes-35.0.0-py2.py3-none-any.whl (2.0 MB)
 2.0/2.0 MB 68.5 MB/s eta 0:00:00
Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (17.2 MB)
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Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl (20 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.40.0-py3-none-any.whl (18 kB)
Downloading opentelemetry_proto-1.40.0-py3-none-any.whl (72 kB)
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Downloading opentelemetry_sdk-1.40.0-py3-none-any.whl (141 kB)
 142.0/142.0 kB 12.8 MB/s eta 0:00:00
Downloading opentelemetry_api-1.40.0-py3-none-any.whl (68 kB)
 68.7/68.7 kB 6.2 MB/s eta 0:00:00
Downloading opentelemetry_semantic_conventions-0.61b0-py3-none-any.whl (231 kB)
 231.6/231.6 kB 19.6 MB/s eta 0:00:00
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.w
 71.6/71.6 kB 6.2 MB/s eta 0:00:00
Downloading pypika-0.51.1-py2.py3-none-any.whl (60 kB)
 60.6/60.6 kB 5.4 MB/s eta 0:00:00
Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: pypika, durationpy, pyproject_hooks, pybase64, opentelemetry-proto, bcrypt, opentelemetry-e
  Attempting uninstall: opentelemetry-proto
    Found existing installation: opentelemetry-proto 1.38.0
    Uninstalling opentelemetry-proto-1.38.0:
      Successfully uninstalled opentelemetry-proto-1.38.0
  Attempting uninstall: opentelemetry-exporter-otlp-proto-common
    Found existing installation: opentelemetry-exporter-otlp-proto-common 1.38.0
    Uninstalling opentelemetry-exporter-otlp-proto-common-1.38.0:
      Successfully uninstalled opentelemetry-exporter-otlp-proto-common-1.38.0
  Attempting uninstall: opentelemetry-api
    Found existing installation: opentelemetry-api 1.38.0
    Uninstalling opentelemetry-api-1.38.0:
      Successfully uninstalled opentelemetry-api-1.38.0
  Attempting uninstall: opentelemetry-semantic-conventions
    Found existing installation: opentelemetry-semantic-conventions 0.59b0
    Uninstalling opentelemetry-semantic-conventions-0.59b0:
      Successfully uninstalled opentelemetry-semantic-conventions-0.59b0
  Attempting uninstall: opentelemetry-sdk
    Found existing installation: opentelemetry-sdk 1.38.0
    Uninstalling opentelemetry-sdk-1.38.0:
      Successfully uninstalled opentelemetry-sdk-1.38.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
google-adk 1.28.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.0 which is incompatible.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have open
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.40.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 wh
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.4
Successfully installed bcrypt-5.0.0 build-1.4.2 chromadb-1.5.6 durationpy-0.10 kubernetes-35.0.0 onnxruntime-1.24.4 opente
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```
# Create ChromaDB client
client = chromadb.Client()

# Create collection
collection = client.create_collection(name="rag_collection")
```

```
# Load pre-trained embedding model
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
modules.json: 100% 349/349 [00:00<00:00, 34.1kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 7.64kB/s]

README.md: 10.5k/? [00:00<00:00, 830kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 5.00kB/s]

config.json: 100% 612/612 [00:00<00:00, 62.8kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 208MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 492.96it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 31.5kB/s]

vocab.txt: 232k/? [00:00<00:00, 2.46MB/s]

tokenizer.json: 466k/? [00:00<00:00, 8.33MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 5.09kB/s]

config.json: 100% 190/190 [00:00<00:00, 17.0kB/s]

```

```

# Sample documents
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

# Unique IDs
ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]

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# Convert documents into embeddings
embeddings = model.encode(documents)

```

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# Add data to collection
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

```

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# Query
query = "What is AI?"

# Convert query to embedding
query_embedding = model.encode([query])

# Search similar documents
results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)

print(results)

```

```
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence']]}
```

```

# Print only retrieved documents
for doc in results['documents'][0]:
    print(doc)

```

Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.

```
# Define retrieval function
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding,
        n_results=2
    )
    return results['documents'][0]
```

```
# Test function
print(retrieve("Explain machine learning"))
```

```
['Machine Learning is a subset of AI.', 'Deep Learning uses neural networks.']
```

Start coding or [generate](#) with AI.